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Labor Supply Distortions of Social Security:

$$V = \ln c_t^Y + \beta [\ln c_{t+1}^O + \gamma \ln (1 - z_{t+1})]$$

s.t.

$$c_t^Y = w_t - s_t$$

$$c_{t+1}^O = (1+r_{t+1})s_t + w_{t+1}z_{t+1}$$

$$\Rightarrow c_t^Y + \frac{c_{t+1}^O}{1+r_{t+1}} = w_t + \frac{w_{t+1}z_{t+1}}{1+r_{t+1}}$$

FOC:

$$c_t: \frac{1}{c_t^Y} = \lambda_t; \quad c_{t+1}^O: \frac{\beta}{c_{t+1}^O} = \frac{\lambda_t}{1+r_{t+1}} \Rightarrow c_{t+1}^O = \beta (1+r_{t+1}) c_t^Y$$

$$z_{t+1}: \frac{\beta \gamma}{1 - z_{t+1}} = \lambda_t \frac{w_{t+1}}{1+r_{t+1}} \Rightarrow z_{t+1} = 1 - \beta \gamma c_t^Y \frac{1+r_{t+1}}{w_{t+1}}$$

$$c_t^Y + \frac{\beta (1+r_{t+1}) c_t^Y}{1+r_{t+1}} = w_t + \frac{w_{t+1}}{1+r_{t+1}} (1 - \beta \gamma c_t^Y \frac{1+r_{t+1}}{w_{t+1}})$$

$$(1+\beta) c_t^Y = w_t + \frac{w_{t+1}}{1+r_{t+1}} - \beta \gamma c_t^Y$$

$$((1+\beta) + \beta \gamma) c_t^Y = w_t + \frac{w_{t+1}}{1+r_{t+1}} \Rightarrow c_t^Y = \frac{1}{1+\beta+\beta \gamma} \left(w_t + \frac{w_{t+1}}{1+r_{t+1}} \right)$$

$$z_{t+1} = 1 - \beta\gamma \frac{1+r_{t+1}}{w_{t+1}} \left[\frac{1}{1+\beta+\beta\gamma} \left(w_t + \frac{w_{t+1}}{1+r_{t+1}} \right) \right]$$

$$= \frac{1 - \beta\gamma}{1 + \beta + \beta\gamma} \left[\frac{1+r_{t+1}}{w_{t+1}/w_t} + 1 \right]$$

$$\text{leisure}_t = \frac{\beta\gamma}{1 + \beta + \beta\gamma} \left[\frac{1+r_{t+1}}{w_{t+1}/w_t} + 1 \right]$$

↳ check by setting $\gamma = 0$, $z_{t+1} = 1$

↳ positive wage growth

$$s_t = w_t - c_t^y = w_t - \frac{1}{1+\beta+\beta\gamma} \left(w_t + \frac{w_{t+1}}{1+r_{t+1}} \right)$$

$$= w_t \left(1 - \frac{1}{1+\beta+\beta\gamma} \right) - \frac{1}{1+\beta+\beta\gamma} \frac{w_{t+1}}{1+r_{t+1}}$$

$$s_t = \frac{1}{1+\beta+\beta\gamma} \left[\beta(1+\gamma)w_t - \frac{w_{t+1}}{1+r_{t+1}} \right]$$

Production Sector:

$$Y_t = AK_t^\alpha N_t^{1-\alpha}$$

$$k_t = \frac{K_t}{N_t}$$

Profit of Firm

$$\Pi_t = Y_t - r_t K_t - \delta K_t - w_t N_t \rightarrow \max_{K_t, N_t}$$

↳ Assume $\delta = 1$

↳ Solve C.E. when reviewing

$$1 + r_t = f'(k_t) = \alpha A k_t^{\alpha-1}$$

$$w_t = f(k_t) - k_t f'(k_t) = (1-\alpha) A k_t^\alpha$$

MCC:

$$\text{Labor: } N_t = \underbrace{L_t}_{\text{young cohort}} + \underbrace{L_{t-1} z_t}_{\text{old cohort}} \Rightarrow \frac{N_t}{L_{t-1}} = 1 + n + z_t$$

$$\text{Capital: } K_{t+1} = L_t s_t \Rightarrow \frac{1}{1+n+z_{t+1}} s_t = k_{t+1}$$

↳ divide by N_{t+1}

take MCC $\Rightarrow$ plug in solved s, z , factor prices, find w, r

Add Social Security:

$$V = \ln c_t^Y + \beta [\ln c_{t+1}^o + \gamma \ln(1 - z_{t+1})]$$

s.t.

$$c_t^Y = w_t(1 - \tau_t) - s_t$$

$$c_t^o = (1 + r_{t+1})s_t + w_{t+1}(1 - \tau_{t+1}) + \underbrace{p_{t+1}(1 - z_{t+1})}_{\substack{\text{pension} \\ \text{length of} \\ \text{retirement}}}$$

Lifetime BC

$$c_t^Y + \frac{c_{t+1}^o}{1 + r_{t+1}} = \underbrace{w_t(1 - \tau_t)}_{\tilde{w}_t} + \underbrace{\frac{w_{t+1}(1 - \tau_{t+1})z_{t+1}}{1 + r_{t+1}}}_{\tilde{w}_{t+1}} + \frac{p_{t+1}(1 - z_{t+1})}{1 + r_{t+1}} \Big|_t$$

$$c_{t+1}^o = \beta(1 + r_{t+1})c_t^Y \quad (\text{same as before})$$

$$\frac{\beta\gamma}{1 - z_{t+1}} = \Big|_t = \frac{\tilde{w}_{t+1} - p_{t+1}}{c_t^Y(1 + r_{t+1})}$$

$$\Rightarrow 1 - z_{t+1} = \beta\gamma c_t^Y \frac{1 + r_{t+1}}{\tilde{w}_{t+1} - p_{t+1}} \quad (\text{plug into LTBC})$$

$$c_t^Y(1 + \beta) = \tilde{w}_t + \frac{p_{t+1}}{1 + r_{t+1}} + \frac{\tilde{w}_{t+1} - p_{t+1}}{1 + r_{t+1}} \left(1 - \beta\gamma c_t^Y \frac{1 + r_{t+1}}{\tilde{w}_{t+1} - p_{t+1}} \right)$$

$\Rightarrow$

$$= \tilde{w}_t + \frac{p_t}{1+r_{t+1}} + \frac{\tilde{w}_{t+1} - p_{t+1}}{1+r_{t+1}} - \beta\gamma c_t^y$$

$$\Rightarrow c_t^y = \frac{1}{1+\beta+\beta\gamma} \left[\tilde{w}_t + \frac{\tilde{w}_{t+1}}{1+r_{t+1}} \right]$$

↳ same as before, just w/ $\tilde{w}$

$$1 - z_{t+1} = \beta\gamma \frac{1+r_{t+1}}{\tilde{w}_{t+1} - p_{t+1}} \cdot \frac{1}{1+\beta+\beta\gamma} \left[\tilde{w}_t + \frac{\tilde{w}_{t+1}}{1+r_{t+1}} \right]$$

$$= \frac{\beta\gamma}{1+\beta+\beta\gamma} \frac{1+r_{t+1}^{ss}}{\tilde{w}_{t+1}^{ss} - p_{t+1}} \left[\tilde{w}_t^{ss} + \frac{\tilde{w}_{t+1}^{ss}}{1+r_{t+1}^{ss}} \right] = l_t^{ss}$$

$$l_t^{no} = \frac{\beta\gamma}{1+\beta+\beta\gamma} \frac{1+r_{t+1}^{no}}{w_{t+1}^{no}} \left[w_t^{no} + \frac{w_{t+1}^{no}}{1+r_{t+1}^{no}} \right]$$

steady state:

$$\frac{l^{no}}{l^{ss}} = \frac{1+r^{no}}{1+r^{ss}} \cdot \frac{w^{ss}(1-\gamma) - p}{w^{no}} \cdot \frac{w^{no} \left(1 + \frac{1}{1+r^{no}} \right)}{w^{ss}(1-\gamma) \left(1 + \frac{1}{r^{ss}} \right)}$$

$$= \frac{1+r^{no}}{1+r^{ss}} \cdot \frac{w^{ss}(1-\gamma) - p}{w^{ss}(1-\gamma)} \cdot \frac{\frac{1}{1+r^{no}}}{\frac{1}{1+r^{ss}}} \cdot \frac{(2+r^{no})}{(2+r^{ss})}$$

$$= \left(1 - \frac{p}{w^{ss}(1-\gamma)} \right) \frac{2+r^{no}}{2+r^{ss}}$$

Generally, $r^s > r^{no}$

leisure higher w/ social security

Ricardian Equivalence:

Cons.

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$\text{s.t.} \quad \sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} - \sum_{t=0}^{\infty} \frac{\tau_t}{(1+r)^t}$$

gov:

$$g_t + B_t(1+r) = B_{t+1} + \tau_t \Rightarrow \sum_{t=0}^{\infty} \frac{g_t}{(1+r)^t} = \sum_{t=0}^{\infty} \frac{\tau_t}{(1+r)^t}$$

OLG:

$$\max_{c_t^y, c_{t+1}^o, s_t} u(c_t^y) + \beta u(c_{t+1}^o)$$

s.t.

$$c_t^y = y - \tau_t - s_t$$

$$c_{t+1}^o = y - \tau_{t+1} + s_t(1+r_{t+1})$$

OLG:

$$\max_{c_t^y, c_{t+1}^o, s_t} u(c_t^y) + \beta u(c_{t+1}^o)$$

s.t.

$$c_t^y = y - \tau_t - s_t$$

$$c_{t+1}^o = y^o - \tau_{t+1} + s_t(1+r_{t+1})$$

LTBC:

$$c_t^y + \frac{c_{t+1}^o}{1+r_{t+1}} = y^y + \frac{y^o}{1+r_{t+1}} - \left(\tau_t + \frac{\tau_{t+1}}{1+r_{t+1}} \right)$$

Because finite life, Ricardian equivalence holds if tax scheme changes effect one cohort. i.e. raise in t , lower in $t+1$.

OLG w/ Bequests

$$c_t^y = w_t - s_t$$

$$c_{t+1}^o = s_t(1+r_{t+1}) - s_{t+1}$$